PROJECT AND TEAM INFORMATION

Project Title

Library Management System

Student / Team Information

Team Name: Library Management Team	
Team Member 1 (Team Lead) Name, Student ID, Roll No, Email	Varun Panwar Student ID: 2510012520 Roll No: 2028262 Email: 2510012520@geu.ac.in 
Team Member 2 Name, Student ID, Roll No, Email	Shubh Saini Student ID: 2510011685 Roll No: 2028159 Email: 2510011685@geu.ac.in 
Team Member 3 Name, Student ID, Roll No, Email	Nilesh Yadav Student ID: 2510370404 Roll No: 2027945 Email: 2510370404@geu.ac.in 

PROPOSAL DESCRIPTION

Motivation

Managing a library manually can become difficult when the number of books, students, and transactions increases. Keeping records of available books, issued books, returned books, due dates, and member information in separate registers can take time and may lead to errors.

Our project, Library Management System, is designed to bring these activities into one application. It will provide book management, member management, issue and return records, book searching, and due-date tracking. Data Structures such as arrays, linked lists, queues, stacks, trees, and hashing can be used wherever they are suitable, while C++ classes and OOP principles will keep the program organized.

This project will also help us understand how Data Structures and Object-Oriented Programming concepts can be applied to a practical library application.

State of the Art / Current Solution

Many libraries use computerized systems to maintain book catalogs and member records. Such systems can search books, record issue and return transactions, calculate due information, and maintain the current availability of books.

In smaller libraries, records may still be maintained manually or through basic spreadsheets. These methods can make searching, updating, and tracking transactions slower as the collection grows.

The proposed Library Management System will combine the main library operations into one application. The project will focus on implementing the core data handling and searching operations using C++ Data Structures and OOP concepts so that their practical use can be understood.

PROJECT GOALS AND MILESTONES

Project Goals

1. Develop a C++ based Library Management System for maintaining books and library members.

2. Implement operations for adding, updating, deleting, and displaying book records.

3. Provide book search using suitable Data Structures for efficient retrieval.

4. Manage book issue and return transactions while maintaining book availability.

5. Use OOP concepts such as classes, objects, encapsulation, constructors, and member functions.

6. Apply Data Structures such as linked lists, queues, stacks, trees, or hashing where appropriate.

Project Milestones

1. Finalize project requirements, features, and overall system design.

2. Design the book, member, and transaction data models in C++.

3. Implement basic book and member management operations.

4. Implement searching and sorting of library records using suitable Data Structures.

5. Implement book issue, return, due-date, and availability management.

6. Integrate all modules, test the complete system, and improve usability and performance.

Project Approach

We will design the Library Management System incrementally in C++. First, we will identify the main entities and divide the program into modules such as Book Management, Member Management, Search, Issue and Return, and Report Management.

Classes such as Book, Member, Transaction, Library, and LibraryManager will be used to represent the major parts of the system. OOP principles will be used to keep the program modular and easier to maintain.

For storing and processing records, appropriate Data Structures will be selected according to the operation. Linked lists can be used for dynamic records, queues can support waiting requests, stacks can support selected history operations, trees can support ordered searching, and hashing can support fast lookup.

Initially, each module will be developed and tested separately. The modules will then be integrated into a single application. Testing will focus on correct record management, search performance, issue and return accuracy, and overall usability.

Project Phases

Phase 1:

Planning & Requirement Analysis

Define the problem statement, objectives, scope, requirements, and select suitable C++/Data Structures.

Phase 2:

System Design & Development

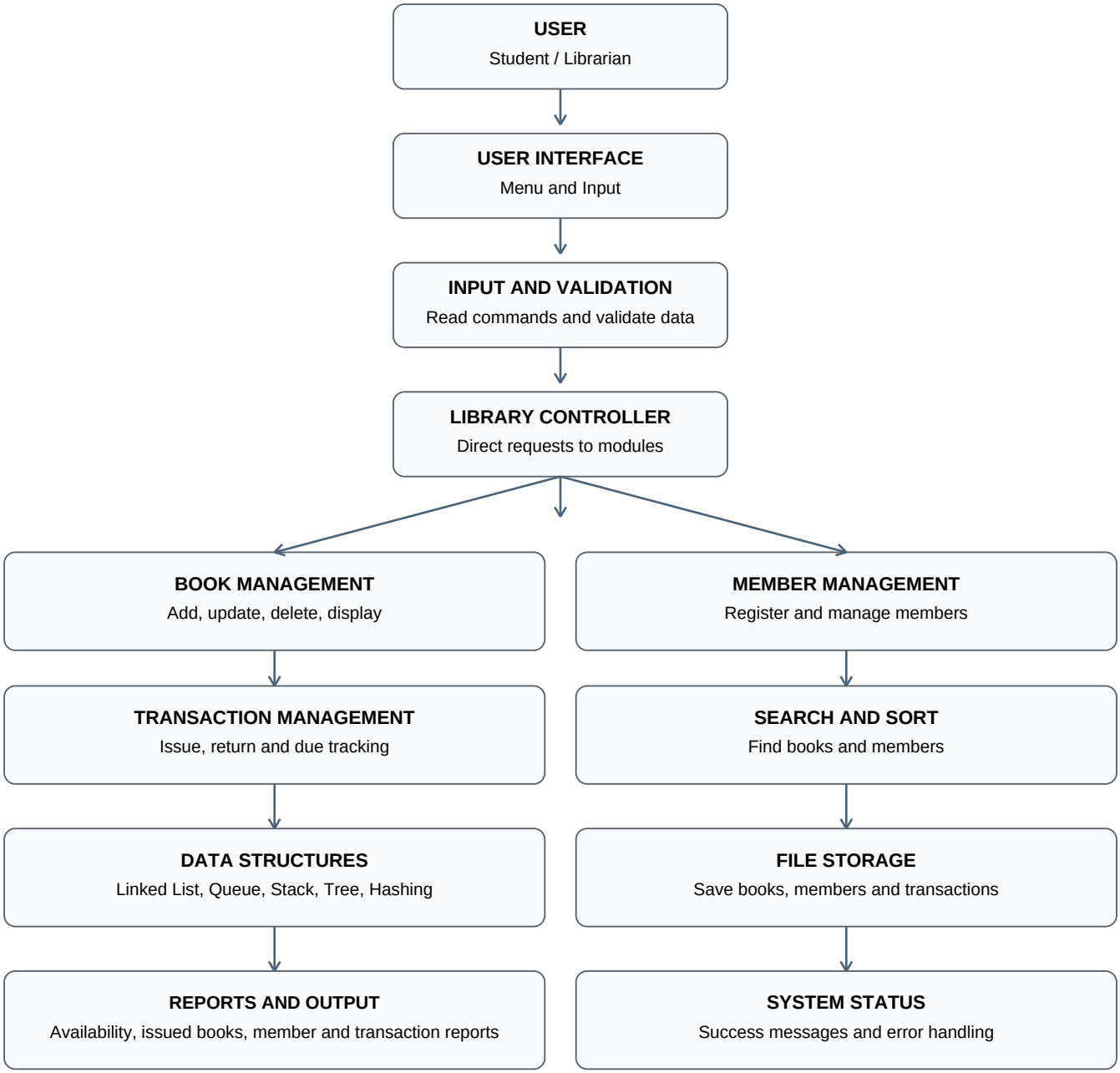
Design system architecture, modules, data handling, and implement library operations using C++ and Data Structures.

Phase 3:

Testing & Final Implementation

Test different library records and transactions, fix errors, verify operations, generate reports, and prepare final documentation/demo.

SYSTEM ARCHITECTURE (HIGH LEVEL DIAGRAM)



Project Outcome / Deliverables

The primary objective of the project is the development of a C++ Library Management System that integrates book management, member management, searching, issue and return processing, availability tracking, and basic reporting into one application.

The application will provide a simple menu-driven interface through which a librarian or authorized user can perform library operations. The system will maintain records in an organized manner and demonstrate the practical use of Data Structures and OOP concepts.

Assumptions

1. The system is intended for a small or medium-sized library.
2. Users will enter valid basic information such as book ID, title, author, member ID, and transaction details.
3. Library records will be stored locally for the project demonstration.
4. The first version will concentrate on book management, member management, search, issue, return, and basic reports.

REFERENCES

1. M. T. Goodrich, R. Tamassia, and M. H. Goldwasser, Data Structures and Algorithms in C++.

2. Robert Lafore, Object-Oriented Programming in C++.

3. Bjarne Stroustrup, Programming: Principles and Practice Using C++.

4. Seymour Lipschutz, Data Structures, Schaum's Outline Series.

PROJECT SUMMARY

Library Management System is a Data Structures and OOP based C++ project intended to simplify common library tasks. The system will maintain book and member information, support searching, manage issue and return transactions, and generate useful status information. The project demonstrates how structured data handling and object-oriented design can be combined to solve a practical problem.